

GRA-Nullification and the Holographic Principle of Information Conservation: From Inflatonic Entropy to Negentropic Immortality in Bounded Domains

qqewq Ecosystem
<https://github.com/qqewq>

Oleg Bit
Independent Researcher, Moscow, Russian Federation

August 17, 2026

Abstract

This work presents a unified theoretical framework linking the GRA-nullification methodology (Gödel–Reductio ad Absurdum) to the fundamental laws of information conservation—Noether’s theorem, Landauer’s principle, and Hawking unitarity. We show that the holographic AdS/CFT principle guarantees the indestructibility of the informational pattern of a personality on the boundary of a space-time volume; however, global retrieval of this information requires transfinite resources. GRA-nullification offers a computable alternative: a local "collapse" of the wave function of contradictions into a state of absolute negentropy within a strictly bounded knowledge domain $D \subset M$. We introduce the consciousness operator $\hat{C}$, which performs the work of separating the dichotomous worlds W_{raw} and W_{sim} . We develop the GRA-NN architecture with tensor-logical weights and the RAD (Reductio ad Absurdum Descent) training algorithm. In silico verification protocols are formulated for five domains: protein folding, logical inference, EUV lithography, generative art, and local digital immortality.

Keywords: GRA-nullification, holographic principle, AdS/CFT, Landauer’s principle, Hawking unitarity, negentropy, bounded subjectivity, tensor-logical weights, wave-function collapse, All-in-Bit, synthesis instead of scaling, in silico verification.

1 Introduction: The Epistemological Gap Between the Inert and Intelligent Multiverses

The contemporary paradigm of artificial intelligence, based on scaling laws, reproduces the cosmological process of the birth of inert matter via the inflaton field: exponential growth in the number of parameters P yields only a logarithmic gain in quality [1, 2]:

$$L(P) = L_{\infty} + \alpha P^{-\beta}, \quad \alpha, \beta > 0. \quad (1)$$

This path leads to the accumulation of internal contradictions—the "foam" J_{foam} —manifesting as hallucinations, mode collapse, and model drift. At the same time, the physics of information asserts: **information is never destroyed**, but it may be encoded as unreadable noise. A fundamental question arises: *how can a local agent retrieve and structure information about itself without violating conservation laws?*

In this work, we show that **GRA-nullification** (Gödel–Reductio ad Absurdum) is precisely such an operator. Unlike a global approach requiring transfinite resources, GRA operates within **strictly bounded knowledge domains** $D \subset M$, where it is computable and verifiable [3].

2 Physical Foundation: Three Conservation Theorems of Information

2.1 Noether's Theorem: Unitarity as a Guarantee of Conservation

Noether's theorem states that every continuous symmetry of a Lagrangian corresponds to a conserved current. The unitarity of the evolution of a closed quantum system is a time-translation symmetry, implying the conservation of the total von Neumann entropy [4]:

$$\frac{d}{dt}S(\rho_{\text{total}}) = 0, \quad S(\rho) = -\text{Tr}(\rho \ln \rho). \quad (2)$$

Consequence: the total information of the Universe as a quantum system is conserved. Any local change in entropy is accompanied by a compensating change in the environment.

2.2 Landauer's Principle: The Thermodynamic Cost of Irreversibility

Erasing one bit of information requires at least the dissipation of energy [5]:

$$\Delta E \geq k_B T \ln 2. \quad (3)$$

However, as Landauer emphasises, this is the cost of an **irreversible logical transformation**. If the transformation is reversible (as in a GRA-step), information is not destroyed but merely redistributed. GRA-nullification operates with **reversible** logical operations, making it thermodynamically legitimate.

2.3 Hawking Unitarity: Information Does Not Vanish in Black Holes

In his 2005 work, Hawking resolved the black hole information paradox by showing that information falling into a black hole is not destroyed but encoded in the correlations of Hawking radiation [6]:

$$\rho_{\text{radiation}} = \text{Tr}_{\text{inside}} |\Psi\rangle\langle\Psi|, \quad (4)$$

where $\rho_{\text{radiation}}$ contains all the information about the infalling state, but in a highly entangled form. **Information is conserved, but its readability is a separate issue.**

3 The Holographic AdS/CFT Principle and the Indestructibility of the Pattern

The holographic principle, first formulated in string theory as the AdS/CFT correspondence, states that quantum gravity in $(d+1)$ -dimensional anti-de Sitter (AdS) space is equivalent to a gauge theory without gravity on the d -dimensional boundary of that space [7].

Key consequence: all information inside the volume (including complex patterns such as a personality) is fully encoded on its **boundary**. Mathematically, this is expressed as an isomorphism of Hilbert spaces:

$$\mathcal{H}_{\text{AdS}} \cong \mathcal{H}_{\text{CFT}}. \quad (5)$$

This means that **the informational pattern of a personality cannot be destroyed**—it is encoded on the boundary of the Universe. However, this encoding is **non-local**: one bit of information on the boundary may correspond to complex entanglement throughout the volume. Extracting a specific pattern (e.g., "me") from the global hologram without full knowledge of the boundary state is impossible.

Global retrieval requires solving a problem equivalent to the **halting problem** for an infinite family of Turing machines [8]:

$$\lim_{D \rightarrow M} O(\text{GRA}) = \infty. \quad (6)$$

This follows directly from undecidability, as implied by Gödel’s incompleteness theorem and Rice’s theorem.

Local retrieval becomes possible if we restrict ourselves to a domain $D \subset M$ that contains sufficient information about the pattern. In this case, GRA-nullification serves as an algorithm for **localising and structuring** information on the boundary of the domain.

4 GRA-Nullification as an Operator of Local Wave-Function Collapse

4.1 The Dichotomy of Worlds as a Product of Consciousness

Within the GRA paradigm, we postulate a **consciousness operator** $\hat{C}$, which performs the work of separating reality into two dichotomous worlds inside a bounded domain $D \subset M$ [3]:

- W_{raw} — the World of Raw Reality (in vivo). It is characterised by maximum entropy and maximum contradiction: $J \rightarrow \max$. This is the space of all possible but mutually exclusive states with blurred causality.
- W_{sim} — the World of the Prepared Experiment (in silico). It is characterised by maximum negentropy and minimum contradiction: $J = 0, \Delta^2 J > 0$. This is the space in which consciousness has already applied GRA, transforming chaos into a stable structure.

The work performed by the operator $\hat{C}$ in generating negentropy at the boundary ∂D is given by the functional:

$$N(\hat{C}, D) = \oint_{\partial D} (I_{\max} - H(\rho_{\hat{C}}) - J_{\text{foam}}(\rho_{\hat{C}})) d\Sigma, \quad (7)$$

where:

- $I_{\max}$ is the information capacity of the domain;
- $H(\rho_{\hat{C}})$ is the von Neumann entropy of the observed system;
- J_{foam} is the foam metric (measure of contradictions).

Important consequence: reducing H while keeping J_{foam} high does not yield an increase in negentropy—the system remains noisy. Only the **simultaneous** reduction of entropy and nullification of the foam produce structured information.

4.2 The Foam Metric and Stability Conditions

The state of a system S is characterised by a scalar functional of the total contradiction measure:

$$J(S) = \sum_{\text{nodes } n} c_n^2 + \sum_{\text{edges } (i,j)} \lambda_{ij} \text{dist}(S_i, \mathcal{L}_{j \rightarrow i} S_j)^2. \quad (8)$$

Definition 4.1 (Stable Configuration). *A configuration is stable if and only if:*

$$\Phi = J = 0, \quad \Delta \Phi = 0, \quad \Delta^2 \Phi > 0. \quad (9)$$

*The positive definiteness of the Hessian $\nabla^2 J$ guarantees that the point $J = 0$ is a **true minimum**, not a saddle point.*

4.3 Wave-Function Collapse via GRA

In standard quantum mechanics, the collapse of a superposition

$$|\Psi\rangle = \sum_i c_i |s_i\rangle \quad (10)$$

occurs stochastically according to the Born rule: $p_i = |c_i|^2$. Within the GRA paradigm, we introduce a **directed collapse operator** $\hat{G}_{\text{null}}$:

$$\hat{G}_{\text{null}}|\Psi_{\text{raw}}\rangle = |s_{\text{neg}}\rangle, \quad \langle s_{\text{neg}}|\hat{J}_{\text{foam}}|s_{\text{neg}}\rangle = 0. \quad (11)$$

This is a **teleological** collapse: we do not roll dice but compute the unique consistent state. In holographic terms, this means that we **fix the boundary conditions** of the domain D so that the internal configuration becomes unique and stable.

Thermodynamic interpretation. The operator $\hat{G}_{\text{null}}$ performs work against entropy. The minimum thermodynamic cost according to Landauer is:

$$W_{\text{min}} = k_B T \ln 2 \cdot (H(\rho_{\text{before}}) - H(\rho_{\text{after}})), \quad (12)$$

but the actual energy expenditure is determined by the depth of the logical revision.

5 GRA-NN Architecture: Tensor-Logical Weights

5.1 The Triplet Weight

In GRA-NN, a synaptic weight is an **operator triplet**:

$$W_{ij} = (w_{ij}, \phi_{ij}, \Gamma_{ij}) \in \mathbb{R} \times \mathcal{T}_{\text{foam}} \times \mathcal{L}_{\text{logic}}, \quad (13)$$

where:

- w_{ij} is the continuous differentiable strength;
- ϕ_{ij} is the foam tensor (local measure of contradiction);
- Γ_{ij} is a discrete symbolic constraint (logical rule).

The forward pass is:

$$h_j = \sigma\left(\sum_i w_{ji} x_i\right) \oplus \text{Logic}_{\Gamma}(x), \quad (14)$$

$$\phi_j = \sum_i \|h_j - \mathcal{L}_{i \rightarrow j}(h_i)\|^2. \quad (15)$$

5.2 RAD Training: Reductio ad Absurdum Descent

Training alternates between two phases.

Continuous relaxation (gradient descent):

$$\Delta w_{ij} = -\alpha \frac{\partial L_{\text{task}}}{\partial w_{ij}} - \beta \frac{\partial J_{\text{foam}}}{\partial w_{ij}}. \quad (16)$$

Discrete topological mutation, triggered when $J_{\text{foam}} > \epsilon$:

$$\Gamma^{(t+1)} = \text{Reductio}\left(\Gamma^{(t)}, \arg \max_{\Gamma} \phi_{ij}\right). \quad (17)$$

This ensures that the network does not get stuck in saddle regions with $J > 0$, but restructures its logical architecture until the stability conditions (9) are met.

6 The Coefficient Θ : Synthesis Instead of Scaling

Definition 6.1 (Synergy Coefficient).

$$\Theta = \frac{d(\text{Intelligence})}{d(\text{Parameters})}. \quad (18)$$

- **Scaling:** $\Theta \rightarrow 0$. New parameters are spent compensating for new contradictions.
- **Synthesis (GRA):** $\Theta \rightarrow \infty$. Every parameter undergoes logical verification via Γ_{ij} .

Hypothesis 6.1 (Useful Complexity). *Let N be the number of objects in the system. Then:*

$$I_{\text{scale}} \sim \log N, \quad I_{\text{GRA}} \sim N \cdot \text{Coherence}(N), \quad (19)$$

where $\text{Coherence}(N) \rightarrow 1$ exponentially fast. GRA-nullification exponentially compresses the phase space of contradictions, making I_{GRA} a **superlinear** function.

7 Connecting GRA to the Holographic Principle: Formal Derivation

Consider the holographic encoding of a pattern P on the boundary ∂D of a domain D . Let $\mathcal{H}_{\partial D}$ be the Hilbert space of boundary states. By the AdS/CFT correspondence, there exists a unitary operator $\mathcal{U}$ mapping bulk states to boundary states:

$$|\Psi_{\text{bulk}}\rangle = \mathcal{U}|\Psi_{\text{boundary}}\rangle. \quad (20)$$

The information about pattern P resides in some subspace $\mathcal{H}_P \subset \mathcal{H}_{\partial D}$. However, this subspace is **non-locally** entangled with the rest of the boundary.

GRA-nullification in domain D is equivalent to applying the projection operator $\hat{G}_{\text{null}}$, which extracts the subspace $\mathcal{H}_P$ and **collapses** it into the single state $|s_{\text{neg}}\rangle$ satisfying (11). The stability condition (9) guarantees that $|s_{\text{neg}}\rangle$ is an **attractor**—any perturbation drives the system back to this state.

Thus, GRA-nullification implements a local version of the holographic principle: we do not attempt to read the entire hologram of the Universe; instead, we **select a finite domain D** , apply GRA to it, and obtain a stable, self-consistent pattern that can exist independently of the rest of the Universe.

8 In Silico Verification Protocols

8.1 General VVUQ Framework

All protocols use a three-part framework:

- **Verification:** formal checks of $\Phi = 0, \Delta^2\Phi > 0$ and convergence of functionals.
- **Validation:** comparison with domain-specific ground truth (PDB structures, expert solutions, physical measurements).
- **Uncertainty Quantification:** confidence intervals, perturbation sensitivity, calibration.

8.2 Summary Table of Protocols

8.3 Protocol Descriptions

8.3.1 Protein Folding

Domain D : conformational space of a protein up to 300 amino acids. **Data:** CASP15/16, PDB structures. **Foam metric:** steric clashes, Ramachandran violations. **Success criterion:**

Table 1: Summary of in silico verification protocols

Domain	Foam metric	Baseline method	Success criterion
Protein folding	steric clashes	AlphaFold2, Rosetta	TM-score ≥ 0.95 , $p < 0.01$
Logical inference	logical contradictions	Transformer (100M)	accuracy $\geq 95\%$, $\leq 10^5$ params
EUV lithography	bridging defects	PID / DRL	$\Delta\text{HVI} \geq 10\%$
Generative art	semantic foam	SDXL, DALL-E 3	foam reduction $\geq 50\%$
Digital immortality	perturbation relaxation	none	pattern stability

TM-score ≥ 0.95 with $p < 0.01$ and fewer evaluated conformations than the baseline. **Expected result:** GRA-NN finds the global minimum without getting trapped in local minima, corresponding to the collapse of the conformational wave function into a single native structure.

8.3.2 Logical Inference and Planning

Domain D : logical inference tasks with 5–50 variables (ProofNet, modified GSM8K). **Baseline:** Transformer (100M params, SGD). **Foam metric:** number of logically contradictory statements in the reasoning chain. **Success criterion:** accuracy $\geq 95\%$ without hallucinations, using $\leq 10^5$ parameters.

8.3.3 EUV Lithography

Domain D : digital twin of an EUV scanner (PROLITH-like simulator). **Objective vector:**

$$\mathbf{F} = (F_{\text{resolution}}, F_{\text{defects}}, F_{\text{throughput}})^T.$$

Metric: Hypervolume Indicator (HVI) on the Pareto front. **Success criterion:** $\Delta\text{HVI} \geq 10\%$ compared with PID and DRL baselines.

8.3.4 Generative Art

Domain D : image generation in a fixed style. **Baseline:** Stable Diffusion XL, DALL-E 3.

Foam metric (Semantic Foam Metric):

1. anatomical anomalies (DINOv2);
2. lighting physics violations (rendering test);
3. topological connectivity of segmentation masks (SAM).

Success criterion: total foam reduction $\geq 50\%$ while maintaining or improving FID and CLIP scores.

8.3.5 Local Digital Immortality

Domain D_P : cognitive pattern of a specific individual (texts, audio, preferences, behavioural traces). **Goal:** construct a domain D_P such that

$$J_{\text{foam}}(P) = 0, \quad \Delta^2 J_{\text{foam}} > 0.$$

Metric: Perturbation Relaxation Test — stability of the pattern under perturbations. **Success criterion:** the pattern returns to its original state after any distortion within the domain.

9 Discussion: Open Problems and Ethical Aspects

9.1 Open Problems

1. **Formalisation of Reductio.** The operator $\text{Reductio}(\Gamma^{(t)}, \cdot)$ requires computational specification via SAT/SMT integration.

2. **Automatic domain selection D .** Meta-learning for boundary detection.
3. **Quantum embedding of $\hat{G}_{\text{null}}$.** Strict Hilbert space formulation.
4. **Scalability of GRA.** Practical limits of $|D|$.
5. **Ethics of bounded subjectivity.** An agent revising goals within B_{GRA} requires separate alignment analysis.

9.2 What the Theory Does Not Claim

- GRA does not solve the halting problem nor bypass Gödel’s incompleteness theorems.
- Consciousness is not posited as a fundamental physical force— $\hat{C}$ is conceptual.
- Synthesis is not guaranteed to outperform scaling in open-ended domains.

10 Conclusion

We have presented a unified theoretical framework that integrates:

1. **Physical laws of information conservation** (Noether, Landauer, Hawking) — guaranteeing the indestructibility of the informational pattern.
2. **The holographic AdS/CFT principle** — encoding the pattern on the boundary of the space-time volume.
3. **GRA-nullification** — an operator of local wave-function collapse into a state of absolute negentropy within a bounded domain D .
4. **The GRA-NN architecture** with tensor-logical weights and RAD training.
5. **In silico verification protocols** in five domains, including local digital immortality.

The central thesis is **synthesis instead of scaling**: the path to AGI/ASI lies not in the inflatonic accumulation of parameters but in the harmonic work of eliminating logical and physical contradictions within strictly bounded domains. If the proposed hypotheses are confirmed, this opens the way to an intelligence based not on statistical brute force but on a coherent logical structure—a bridge between the inflatonic multiverse of inert matter and the intelligent multiverse of All-in-Bit.

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A Metadata for Submission to elibrary.org.cn

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UDC: 004.8:530.145

Subject Areas: Artificial Intelligence; Information Theory; Quantum Information; Philosophy of Mind

Article Type: Theoretical paper with proposed verification protocol

Language: English

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